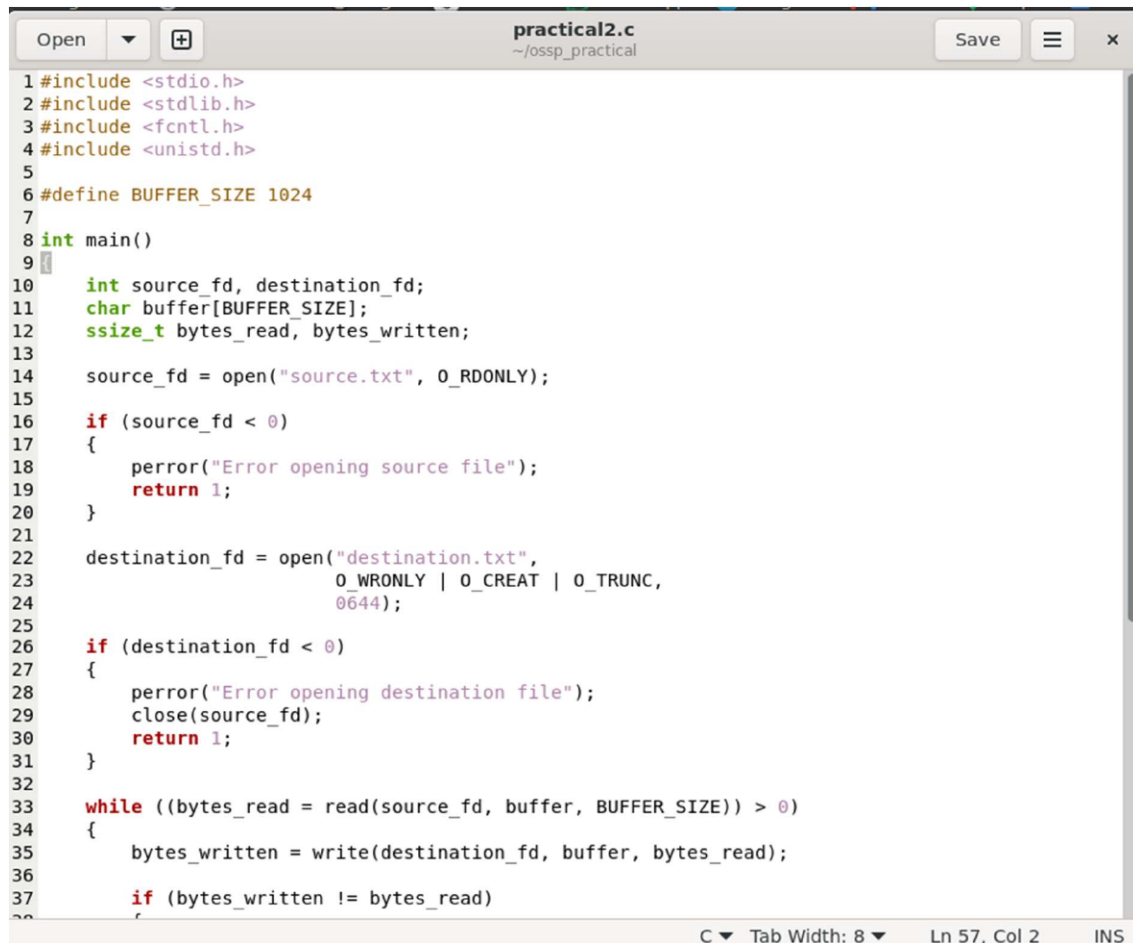
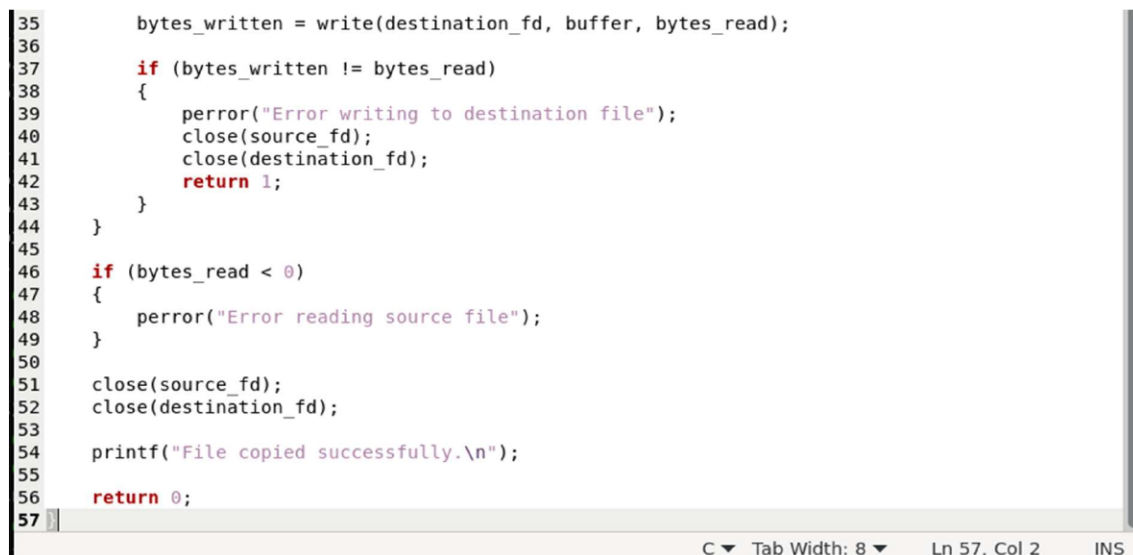


Practical – 2 – OS



The screenshot shows a code editor window titled "practical2.c" with the path "~/ossp_practical". The code includes standard headers, defines a buffer size, and opens source and destination files. It includes error handling for file opening failures.

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <fcntl.h>
4 #include <unistd.h>
5
6 #define BUFFER_SIZE 1024
7
8 int main()
9 {
10     int source_fd, destination_fd;
11     char buffer[BUFFER_SIZE];
12     ssize_t bytes_read, bytes_written;
13
14     source_fd = open("source.txt", O_RDONLY);
15
16     if (source_fd < 0)
17     {
18         perror("Error opening source file");
19         return 1;
20     }
21
22     destination_fd = open("destination.txt",
23                          O_WRONLY | O_CREAT | O_TRUNC,
24                          0644);
25
26     if (destination_fd < 0)
27     {
28         perror("Error opening destination file");
29         close(source_fd);
30         return 1;
31     }
32
33     while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
34     {
35         bytes_written = write(destination_fd, buffer, bytes_read);
36
37         if (bytes_written != bytes_read)
```



The screenshot shows the continuation of the C program, including error handling for writing failures, closing files, and printing a success message.

```
35         bytes_written = write(destination_fd, buffer, bytes_read);
36
37         if (bytes_written != bytes_read)
38         {
39             perror("Error writing to destination file");
40             close(source_fd);
41             close(destination_fd);
42             return 1;
43         }
44     }
45
46     if (bytes_read < 0)
47     {
48         perror("Error reading source file");
49     }
50
51     close(source_fd);
52     close(destination_fd);
53
54     printf("File copied successfully.\n");
55
56     return 0;
57 }
```

Practical – 2 – OS

```
Saketh@SakethVutharavalli:~/ossp_practical$ gedit practical2.c
Saketh@SakethVutharavalli:~/ossp_practical$ echo "This is my OSSP Practical 2 file." > source.txt
Saketh@SakethVutharavalli:~/ossp_practical$ cat source.txt
This is my OSSP Practical 2 file.
Saketh@SakethVutharavalli:~/ossp_practical$ gcc practical2.c -o practical2
Saketh@SakethVutharavalli:~/ossp_practical$ ./practical2
File copied successfully.
Saketh@SakethVutharavalli:~/ossp_practical$ ls
destination.txt practical1 practical1.c practical2 practical2.c source.txt
Saketh@SakethVutharavalli:~/ossp_practical$ cat destination.txt
This is my OSSP Practical 2 file.
Saketh@SakethVutharavalli:~/ossp_practical$
```

Strace Output

```
Saketh@SakethVutharavalli:~/ossp_practical$ grep -E 'openat\(..*sample\.txt|read\(|write\(|close\(|exit_group' trace.txt
close(3) = 0
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\3\0\0\0\1\0\0\0\220\243\2\0\0\0\0"... , 832) = 832
close(3) = 0
read(3, "# Locale name alias data base.\n#"..., 4096) = 2996
read(3, "", 4096) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
close(3) = 0
openat(AT_FDCWD, "sample.txt", 0_RDONLY) = 3
read(3, "Hello OSSP\n", 131072) = 11
write(1, "Hello OSSP\n", 11) = 11
read(3, "", 131072) = 0
close(3) = 0
exit_group(0) = ?
Saketh@SakethVutharavalli:~/ossp_practical$
```